

LUKA, Czech Republic

WMO: 117100

Lat: 49.6522N

Lon: 16.9533E

Elev: 1692

StdP: 13.82

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	13.1	2.1	6.5	11.2	7.0	8.4	14.3	33.5	36.8	29.7	36.0	7.3	40	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	84.1	65.6	80.6	64.7	77.6	63.5	67.8	78.9	66.2	77.2	64.7	75.0	9.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	94.6	72.6	62.3	89.5	70.7	60.9	85.0	69.0	33.1	79.2	31.8	77.4	30.6	75.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	22.8	20.0	4.8	88.8	5.8	4.0	0.6	91.6	-2.8	94.0	-6.0	96.2	-10.3	99.2		
(5)			4.3	70.9	5.8	1.9	0.1	72.2	-3.3	73.3	-6.5	74.4	-10.8	75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	27.8	30.8	37.5	47.7	55.5	61.5	65.4	65.1	56.3	47.1	38.5	29.8	
	DBStd	15.18	8.17	8.43	7.88	8.11	6.81	6.70	6.37	6.15	6.44	7.24	7.41	7.59	
	HDD50	2910	687	537	390	135	25	2	0	0	16	144	348	626	
	HDD65	6786	1152	956	852	520	302	145	76	75	269	554	794	1091	
	CDD50	1824	0	0	3	66	195	349	477	469	205	55	5	0	
	CDD65	223	0	0	0	0	7	41	88	79	8	0	0	0	
	CDH74	1579	0	0	0	3	41	267	667	561	40	0	0	0	
	CDH80	359	0	0	0	0	2	41	176	137	3	0	0	0	
Wind	WSAvg	9.6	9.9	10.4	10.7	9.9	9.6	8.9	9.1	8.4	9.1	9.1	10.0	9.9	
Precipitation	PrecAvg	23.6	1.3	1.1	1.6	1.5	2.5	2.8	3.3	2.9	2.1	1.5	1.6	1.3	
	PrecMax	31.8	2.2	3.1	3.9	3.2	6.1	4.9	10.6	6.0	4.9	3.9	3.4	2.4	
	PrecMin	17.8	0.3	0.2	0.2	0.2	0.9	0.4	0.2	1.1	0.5	0.2	0.0	0.5	
	PrecStd	3.0	0.6	0.7	0.9	0.8	1.1	1.1	2.0	1.4	1.2	0.9	0.7	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	51.7	60.9	72.7	78.2	84.4	89.1	88.6	78.9	67.7	57.9	49.0	
		MCWB	45.0	44.5	48.7	57.1	62.2	66.9	65.4	65.1	61.6	58.7	52.1	45.7	
	2%	DB	45.2	47.7	56.3	68.1	74.4	80.5	84.8	83.7	73.7	63.5	53.9	44.9	
		MCWB	42.0	43.1	46.6	54.3	60.8	65.4	65.8	65.0	60.9	56.4	50.2	42.4	
	5%	DB	41.2	44.6	52.4	64.0	70.9	77.0	81.0	79.8	70.0	60.3	51.1	42.1	
		MCWB	38.8	40.7	44.6	52.2	59.0	63.9	64.7	64.3	59.2	54.7	48.2	40.1	
	10%	DB	38.2	41.4	48.5	60.3	67.3	73.7	77.5	76.5	66.7	57.2	48.3	39.4	
		MCWB	36.4	38.4	42.9	50.3	57.0	61.9	63.6	63.3	57.8	53.1	46.0	38.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	46.2	50.2	58.2	64.6	69.8	70.0	69.3	64.7	60.9	53.1	46.2
			MCDB	48.4	50.3	57.9	69.9	75.1	81.4	81.7	80.5	74.3	64.8	55.8	48.1
		2%	WB	42.2	43.5	47.7	55.5	62.1	67.2	67.9	67.2	62.2	57.4	50.8	42.9
			MCDB	44.5	47.0	54.3	66.1	71.9	77.0	79.7	78.4	70.3	61.5	53.6	44.5
5%		WB	39.0	41.3	45.7	53.2	60.1	65.2	66.1	65.6	60.5	55.5	48.3	40.4	
		MCDB	41.1	44.2	51.2	62.1	68.8	74.5	77.7	76.6	68.0	59.2	50.7	41.9	
10%		WB	36.6	38.7	43.5	51.1	58.1	63.1	64.5	64.1	58.8	53.5	46.3	38.2	
		MCDB	38.1	41.0	47.8	58.9	65.7	71.9	75.0	74.4	65.4	56.9	48.2	39.4	
Mean Daily Temperature Range	5% DB	MDBR	7.5	9.2	11.9	15.3	15.7	16.2	16.9	17.1	13.7	10.9	7.7	6.9	
		MCDBR	10.0	12.0	17.4	19.5	19.6	20.2	21.8	21.6	18.5	14.4	10.3	9.0	
		MCWBR	8.1	8.4	10.4	10.2	10.2	9.6	8.4	8.5	8.6	8.5	7.4	7.4	
	5% WB	MCDBR	9.6	10.9	15.8	18.4	18.3	18.9	20.1	19.4	16.8	13.4	9.4	8.7	
		MCWBR	8.1	8.3	10.3	10.1	10.3	9.9	9.2	8.7	8.7	8.6	7.4	7.5	
Clear-Sky Solar Irradiance	taub		0.296	0.319	0.355	0.400	0.399	0.399	0.412	0.407	0.378	0.353	0.330	0.295	
	taud		2.346	2.332	2.298	2.229	2.266	2.293	2.261	2.290	2.347	2.393	2.401	2.394	
	Ebn at Noon		238	260	269	267	273	273	267	263	258	244	222	222	
	Edh at Noon		21	28	34	41	41	40	41	38	32	26	21	18	
All-Sky Solar Radiation	RadAvg		271	519	869	1338	1599	1716	1637	1469	1038	600	292	206	
	RadStd		47	70	148	159	194	154	175	136	149	90	34	20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.95	N/A	N/A	+1.42	+0.74	+0.65	-220	-288	+133	+54

Nomenclature: See separate page